

Tracing Vehicular Mobility in ns-3

ns-3 is free simulation software that's meant for educational, research and development purposes. It has a number of high quality, tested and validated models, and is backed by a large community of developers and users.



The ns-3 simulator is an open source project meant for educational and development purposes. It is also used to enact discrete-event simulation. Practical implementation or experimental testing of new protocols on real-world networks is not always possible. ns-3 helps to abstract real-world entities and allows to simulate, as well as to compute complex calculations on huge networks of computer systems. It can be treated as a real-time network emulator, as it can simulate scenarios on realistic data. It provides a platform for developers to build simulation models and can comprehend every single step, i.e., the entire workflow of the simulation, from configuration to the collection and analysis of traces.

The ns-3 project includes and supports various real-world protocols and its implementations. It supports various routing protocols, which include OLSR, AODV and many more for IP-based applications on IP networks. It also supports non-IP based network simulation. The ns-3 simulator majorly involves models for Wi-Fi, WiMax and LTE.

Mobility provision in the ns-3 simulator

The ns-3 simulator provides support for mobility in the form of mobility models and helper classes. Mobility models

are used to track and perpetuate the current position and speed of an object. Mobility helper classes are used to place nodes and set up mobility models. Users communicate with the mobility system using the mobility helper classes as they integrate a mobility model and position allocator, and can be used along with a node container to install mobility functions on a set of nodes. The ns-3 simulator provides a ns-2 mobility helper class, which allows it to interpret the ns-2 file format of the trace file. The following statements show the ns-2 trace file format:

```
$Node_ (node number) set X_ x
$Node_ (node number) set Y_ y
$Node_ (node number) set Z_ z
$ns_ at $time $Node_ (node number) setdest x2 y2 speed
$ns_ at $time $Node_ (node number) setdest x3 y3 speed
$ns_ at $time $Node_ (node number) setdest x4 y4 speed
$ns_ at $time $Node_ (node number) setdest x5 y5 speed
```

In the above traces, *x* and *y* are initial positions while *z* specifies the speed in m/s. These are initialised using the set statements. The command *setdest* is used to define the new position (*x2*, *y2*) of the node at the given time (*\$time*), which